

Section : Common

Q.1 The area (in sq. units) of the region bounded by the parabola $y^2 = 8x$ and the line $x = 2$ is

1. $\frac{32}{3}$
2. $\frac{1}{3}$
3. $\frac{16}{3}$
4. $\frac{4}{3}$

Options

1. 1
2. 2
3. 3
4. 4

Q.2 $\int_2^5 |x - 3| dx$ equals

1. $\frac{3}{2}$
2. $\frac{5}{2}$
3. $\frac{1}{2}$
4. 5

Options

1. 1
2. 2
3. 3
4. 4

Q.3 If $y = 5e^{2x} + 4e^{3x}$, then $\frac{d^2y}{dx^2}$ equals:

1. $3(2e^{2x} + 6e^{3x})$
2. $4(5e^{2x} + 9e^{3x})$
3. $5(4e^{2x} + 9e^{3x})$
4. $2(5e^{2x} + 8e^{3x})$

Options

1. 1
2. 2
3. 3
4. 4

Q.4 If A is a square matrix and I is the identity matrix of same order such that $A^2 = I$, then $3(A - I)^3 + 3(A + I)^3 - 15A$ is equal to

1. $19A$
2. $9A$
3. $10A$
4. $5A$

Options

1. 1
2. 2
3. 3
4. 4

Q.5 If $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$, then the matrix AB is equal to

1. $\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$
2. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
3. $\begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$

4. $\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$

Options 1. 1

2. 2

3. 3

4. 4

Q.6 Which of the following are first order linear differential equations?

(A) $\frac{dx}{dy} + P_1(y)x = Q_1(y)$: $P_1(y)$ and $Q_1(y)$ are functions of y or constant functions

(B) $\frac{dy}{dx} + P_2(x)y = Q_2(x)$: $P_2(x)$ and $Q_2(x)$ are functions of x or constant functions

(C) $(x + y) \frac{dy}{dx} = x - 2y$

(D) $(1 + x^2) \frac{dy}{dx} - 2xy = x^2 + 3$

Choose the \textbf{correct} answer from the options given below:

1. (A) and (B) only

2. (A), (B) and (C) only

3. (A), (B) and (D) only

4. (A), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.7 The point of local maxima of the function $f(x) = (x - 2)^5(x + 2)^2$ is

1. 2

2. $-\frac{6}{7}$

3. $\frac{2}{7}$

4. -2

Options 1. 1

2. 2

3. 3

4. 4

Q.8 The probability distribution of a random variable X is given by

X 0 1 2

$P(X)$ k $2k$ $3k$

If $k > 0$, then $P(0 < X \leq 2)$ is equal to

1. $\frac{1}{6}$

2. $\frac{5}{6}$

3. $\frac{7}{8}$

4. $\frac{9}{16}$

Options 1. 1

2. 2

3. 3

4. 4

Q.9 Let $A = [a_{ij}]_{n \times n}$ be a matrix, then match \textbf{List-I} with \textbf{List-II}

\textbf{List-I} \textbf{List-II}

(A) $|A| = 0$ (I) A is a symmetric matrix

(B) $|A| \neq 0$ (II) A is a skew-symmetric matrix

(C) $A^T = A$ (III) A is a singular matrix

(D) $A^T = -A$ (IV) A is a non-singular matrix

Choose the \textbf{correct} answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

3. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.10 $\int \frac{e^{7\log_e x} - e^{6\log_e x}}{e^{4\log_e x} - e^{3\log_e x}} dx$ is equal to:

(Here, c is an arbitrary constant)

1. $-\frac{x}{2} + c$

2. $\frac{x^2}{2} + c$

3. $\frac{x^3}{3} + c$

4. $\frac{x^4}{4} + c$

Options 1. 1

2. 2

3. 3

4. 4

Q.11

If $A = \begin{bmatrix} 0 & 0 & \sqrt{7} \\ 0 & \sqrt{7} & 0 \\ \sqrt{7} & 0 & 0 \end{bmatrix}$, then $|\text{adj } A|$ is equal to

1. 7

2. 49

3. 343

4. 0

Options 1. 1

2. 2

3. 3

4. 4

Q.12 The solution set of the linear constraints $x - 2y \geq 0$, $2x - y \leq -4$, $x \geq 0$ and $y \geq 0$ is

1. ϕ

2. $\{(-1, 0), (0, 2)\}$

3. $\{(-1, 0), (0, 2), (0, 0)\}$

4. unbounded region in first quadrant

Options 1. 1

2. 2

3. 3

4. 4

Q.13 A car is moving along the curve $y = x^3 + 12$. The point(s) on the curve at which the rate of change of its y -coordinate at a certain time is 3 times the rate of change of its x -coordinate is/are

1. (1, 13) only

2. (-1, 12) only

3. (1, 13) and (-1, 11)

4. (-1, 13) only

Options 1. 1

2. 2
3. 3
4. 4

Q.14 The corner points of the bounded feasible region associated with the LPP: Maximize $Z = px + qy$, $p, q > 0$ are $(0, 0)$, $(3.5, 0)$, $\left(\frac{112}{59}, \frac{135}{59}\right)$ and $(0, 3)$. If the optimum value of Z occurs at both $\left(\frac{112}{59}, \frac{135}{59}\right)$ and $(0, 3)$, then

1. $3q = 5p$
2. $8p = 3q$
3. $5p = 8q$
4. $3p = 5q$

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.15 The solution of the differential equation $\log_e \left(\frac{dy}{dx} \right) = 5x + 2y$ is given by

1. $5e^{5x} + 2e^{-2y} + C = 0$: C is an arbitrary constant
2. $2e^{5x} + 5e^{-2y} + C = 0$: C is an arbitrary constant
3. $2e^{-5x} + 5e^{2y} + C = 0$: C is an arbitrary constant
4. $5e^{-5x} + 2e^{2y} + C = 0$: C is an arbitrary constant

- Options
1. 1
 2. 2
 3. 3
 4. 4

Section : Applied Mathematics

Q.16 It is currently 9:40 AM. What will be the time in next 610 hours?

1. 9:50 AM
2. 7:40 AM
3. 9:50 PM
4. 7:40 PM

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.17 If an investment value gets doubled in 10 years then its compound annual growth rate (CAGR) is:
(Given $2^{1/10} = 1.0729$)

1. 5%
2. 10%
3. 12%
4. 7.29%

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.18 $\int \frac{1}{(x+1)(x+2)} dx$ is equal to

1. $\log_e \left| \frac{x+2}{x+1} \right| + C$

2. $\log_e \left| \frac{x+1}{x+2} \right| + C$

3. $\log_e |(x+1)(x+2)| + C$

4. $\log_e |2x+3| + C$

Options 1. 1

2. 2

3. 3

4. 4

Q.19 In reference to Inferential Statistics, for 20 degrees of freedom, the least statistical t-value among the given below is:

1. $t(0.005)$

2. $t(0.01)$

3. $t(0.025)$

4. $t(0.05)$

Options 1. 1

2. 2

3. 3

4. 4

Q.20 Ramesh plans to save some amount required after 10 years for higher studies of his son.

He expects the cost of these studies to be Rs. 1,00,000.

How much should be saved at the beginning of each year to accumulate this amount at the end of 10 years, if the interest rate is 12% compounded annually?

(Given $(1.12)^{11} = 3.477$)

1. Rs. 4029

2. Rs. 5702

3. Rs. 5091

4. Rs. 5901

Options 1. 1

2. 2

3. 3

4. 4

Q.21 Math input error

Using straight line method the annual depreciation of the machine is:

Math input error

Math input error

Math input error

Math input error

Options 1. 1

2. 2

3. 3

4. 4

Q.22 In a 900 meters race, Ram gives Shyam a start of 150 meters and defeats him by 50 seconds. If the speed of Ram is 4.5 m/sec, then the speed of Shyam is

1. 3.5 m/sec
2. 2 m/sec
3. 3 m/sec
4. 4.5 m/sec

Options 1. 1
2. 2
3. 3
4. 4

Q.23 The corner points of the bounded feasible region for an LPP are (0, 20), (3, 12), (6, 8), (0, 15).
The objective function is $Z = ax + \beta y$, where $a, \beta > 0$.
If the maximum of Z occurs at the corner points (3, 12) and (6, 8),
then the relationship between a and β is:

1. $a = \frac{2}{3}\beta$
2. $\beta = \frac{4}{3}a$
3. $a + \beta = \frac{1}{2}$
4. $a = \frac{4}{3}\beta$

Options 1. 1
2. 2
3. 3
4. 4

Q.24 The differential equation representing the family of curves $y = Ax + \frac{B}{x}$, $x \neq 0$,
where A and B are arbitrary constants, is given by

1. $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$
2. $8x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} - y = 0$
3. $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y = 0$
4. $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = 0$

Options 1. 1
2. 2
3. 3
4. 4

Q.25 Consider the following data:

Year(x)	2010	2011	2012	2013	2014
Sale(in crore Rs.)(y)	9	18	21	29	38

A straight line trend by the method of least squares is:

1. $y = 46 + 9.6(x - 2012)$
2. $y = 23 + 6.9(x - 2012)$
3. $y = 23 + 7.5(x - 2012)$
4. $y = 46 + 9.6x$

Options 1. 1
2. 2

3. 3

4. 4

Q.26 A random variable y has the following probability distribution

y 1 2 3 4 5

$P(y)$ $2k$ $3k$ k $4k$ $5k$

Match List-I with List-II

List-I

List-II

(A) $P(y > 2)$

(I) $\frac{2}{5}$

(B) k

(II) $\frac{2}{3}$

(C) $P(y \leq 3)$

(III) $\frac{8}{15}$

(D) $P(2 \leq y \leq 4)$

(IV) $\frac{1}{15}$

Choose the correct answer from the options given below

1. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

2. (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

3. (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

4. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Options 1. 1

2. 2

3. 3

4. 4

Q.27 500 g of jaggery syrup has 30% jaggery in it.

The quantity of jaggery that should be added to make it 50% of syrup is:

1. 200 g

2. 750 g

3. 500 g

4. 250 g

Options 1. 1

2. 2

3. 3

4. 4

Q.28 For the given values 8, 10, 12, 14, 16, the three-year moving averages are:

1. 10, 12, 14

2. 11, 13, 15

3. 16, 14, 18

4. 12, 14, 16

Options 1. 1

2. 2

3. 3

4. 4

Q.29 If $xy + \frac{x^2}{y} = x^3y + y$, then $\frac{dy}{dx}$ is equal to

1. $\frac{y(3x^2y^2 + 2x + y^2)}{x^2y - x^2y^3 + x^2 - y^2}$
2. $\frac{y(3x^2y^2 - 2x - y^2)}{xy^2 - x^3y^2 - x^2 - y^2}$
3. $\frac{y(3x^2y^2 - 2x + y^2)}{x^2y - x^2y^3 + x^2 + y^2}$
4. $\frac{y(3y^2x^2 - y^2 + 2x)}{x^2y^3 - x^2y - x - y^2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Match List-I with List-II

List-I

- (A) The chance of observing a specific outcome in an experiment
- (B) A method used to estimate the population parameters
- (C) A value that describes an entire population
- (D) A measurable characteristic of a sample

List-II

- (I) Parameter
- (II) Estimation
- (III) Statistic
- (IV) Probability distribution

Choose the correct option:

1. (A)-(IV), (B)-(II), (C)-(I), (D)-(III)
2. (A)-(III), (B)-(II), (C)-(IV), (D)-(I)
3. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
4. (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Ravi's speed of rowing in still water is 6 km/hr

The speed of the stream is 3 km/hr

He rows between two points in a river and returns to the same point

He took 30 minutes more to cover the distance upstream than downstream

Find the distance between the two points

1. 3.53 km
2. 2.25 km
3. 5.12 km
4. 1.57 km

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Assume P, Q, R and W are matrices of order $3 \times 3, a \times 4, b \times c$ and $d \times a$ respectively

If $PQ + WR$ is well defined, find the value of $ab + cd$

1. 18
2. 28
3. 21
4. 13

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Two pipes A and B can fill a tank in 5 hours and 6 hours respectively
Pipe C can empty the tank in 12 hours
If all three pipes are opened together, find the time taken to fill the tank

1. $2\frac{3}{4}$ hours
2. 3 hours
3. 2 hours
4. $3\frac{9}{17}$ hours

Options 1. 1
2. 2
3. 3
4. 4

Q.34 If $\mathbb{Z}$ and $\mathbb{R}$ denote set of integers and set of real numbers respectively, then match List I with List II.

List-I

List-II

- | | |
|---|--|
| (A) $5x - 3 < 3x + 1, x \in \mathbb{Z}$ | (I) $x \in (-\infty, -3]$ |
| (B) $3x + 17 \leq 2(1 - x), x \in \mathbb{R}$ | (II) $x \in (-\infty, -1)$ |
| (C) $13x + 17 < 2(1 - x), x \in \mathbb{R}$ | (III) $\{\dots, -4, -3, \dots, 0, 1\}$ |
| (D) $\frac{2x + 3}{5} - 2 > \frac{3(x - 2)}{5}, x \in \mathbb{Z}$ | (IV) $\{\dots, -4, -3, -2\}$ |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
2. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
3. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
4. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)

Options 1. 1
2. 2
3. 3
4. 4

Q.35 If the corner points of the bounded feasible region of an LPP are $(0, 2), (3, 0), (6, 0), (6, 8)$ and $(0, 5)$ then the minimum value of the objective function $F = 4x + 6y$ occurs at

1. $(0, 2)$ only
2. $(3, 0)$ only
3. the midpoint of the line segment joining $(0, 2)$ and $(3, 0)$
4. every point on the line segment joining $(0, 2)$ and $(3, 0)$

Options 1. 1
2. 2

3. 3

4. 4

Q.36

If $A = \begin{bmatrix} a & 1 & -1 \\ 0 & b & 4 \\ 4 & 4 & c \end{bmatrix}$ and $abc = 12, b = 4a$, find the value of $|A(\text{adj}A)|$

1. 28^3

2. 12^3

3. 28^2

4. 48^2

Options

1. 1

2. 2

3. 3

4. 4

Q.37

If $A = \begin{bmatrix} 2 & -2 & 1 \\ 0 & 4 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 & 7 \\ 2 & 0 & 6 \end{bmatrix}$

are two matrices such that $3A - 2B + 4C = O$, then matrix C is equal to:

1. $\begin{bmatrix} 1 & -10 & 11/2 \\ -1 & -3 & 6 \end{bmatrix}$

2. $\begin{bmatrix} -1 & 1/2 & 11/4 \\ 1 & -3 & 3/2 \end{bmatrix}$

3. $\begin{bmatrix} -1 & 2 & 3/2 \\ 1 & 3 & 1/2 \end{bmatrix}$

4. $\begin{bmatrix} -10 & 11/2 & 1 \\ -3 & 6 & -1 \end{bmatrix}$

Options

1. 1

2. 2

3. 3

4. 4

Q.38

The effective rate equivalent to a nominal rate of 12% compounded quarterly is:

(Given $(1.03)^4 = 1.1256$)

1. 14.25%

2. 12.56%

3. 10.36%

4. 12%

Options

1. 1

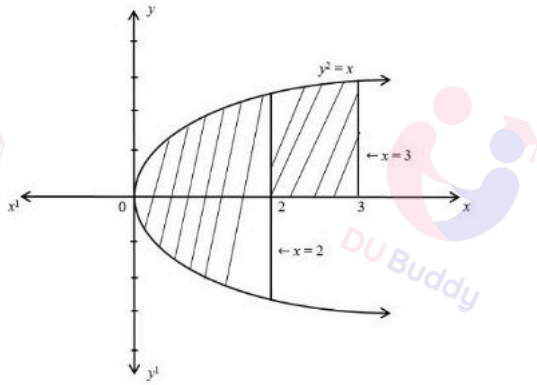
2. 2

3. 3

4. 4

Q.39

Shown below is the graph of parabola $y^2 = x$, the area (in sq. units) of the shaded region is:



1. $2\sqrt{3}$
2. $\frac{4\sqrt{2}}{3} + 2\sqrt{3}$
3. $4\sqrt{3}$
4. $\frac{4\sqrt{2}}{3} - 2\sqrt{3}$

Options

1. 1
2. 2
3. 3
4. 4

Q.40 The demand function (in Rs.) for a product is $P = 20 - 0.25x$ where P is the price per unit and x is the number of units sold. Find the price of one unit when the revenue is maximized.

1. Rs. 20
2. Rs. 10
3. Rs. 30
4. Rs. 40

Options

1. 1
2. 2
3. 3
4. 4

Q.41 A bag contains 6 red balls, 4 green balls and 10 blue balls. Three balls are drawn with replacement. Find the probability of getting at least one green ball.

1. $\frac{1}{4}$
2. $\frac{61}{125}$
3. $\frac{1}{2}$
4. $\frac{29}{57}$

Options

1. 1
2. 2
3. 3
4. 4

Q.42 Increase in the number of patients in the hospitals due to heat stroke is:

1. Secular trend
2. Irregular variation
3. Seasonal variation
4. Cyclic variation

Options

1. 1
2. 2
3. 3
4. 4

Q.43 A random variable X denotes the number of sixes obtained in three throws of a die
Find the mean of the distribution

1. $\frac{1}{3}$

2. $\frac{1}{2}$

3. $\frac{1}{8}$

4. $\frac{2}{3}$

Options

1. 1
2. 2
3. 3
4. 4

Q.44

$$A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -4 & \frac{5}{2} \\ -\frac{1}{2} & 3 & -\frac{3}{2} \\ \frac{1}{2} & y & \frac{1}{2} \end{bmatrix}$$

Find the value of $8x + 5y$

1. 0

2. 1

3. 2

4. 3

Options

1. 1
2. 2
3. 3
4. 4

Q.45 A fair coin is tossed 100 times

Find the probability of getting head an odd number of times

Options:

1. $\frac{1}{2}$

2. $\frac{1}{8}$

3. $\frac{3}{8}$

4. $\frac{80}{100}$

Options 1. 1

2. 2

3. 3

4. 4

Q.46 A man takes a personal loan of Rs. 3,00,000 at 6% per annum

compounded monthly to be repaid by equal monthly installments in 3 years

Find the EMI using flat rate method

Options:

1. Rs. 9833.33

2. Rs. 11,333.3

3. Rs. 17,333.3

4. Rs. 12,833.3

Options 1. 1

2. 2

3. 3

4. 4

Q.47 Let A be a non-singular matrix of order $n \times n$

Then $|\text{adj}(3A)|$ is equal to:

Options:

1. $3|A|^{n-1}$

2. $3^{n(n-1)}|A|^{n-1}$

3. $3^n|A|^{n-1}$

4. $3^n|A|^n$

Options 1. 1

2. 2

3. 3

4. 4

Q.48 In a survey of a sample of 300 individuals, 180 gave response 'Yes',

100 gave response 'No' and 20 gave 'No response'

Find the point estimate of proportion in the population who responded 'Yes'

Options:

1. 0.5
2. 0.2
3. 0.6
4. 0.4

Options 1. 1

2. 2
3. 3
4. 4

Q.49 A piece of machinery is bought for Rs. 50,000

In the first year, it depreciates by 15%

In each subsequent year, the depreciation rate increases by 5%

Find the value of the machinery after 3 years

Options:

1. Rs. 32,700
2. Rs. 24,000
3. Rs. 25,500
4. Rs. 42,000

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Consider $f(x) = \sin(3x) + 4$, $\forall x \in \mathbb{R}$

(A) Maximum value of $f(x)$ is 5

(B) Minimum value of $f(x)$ is 3

(C) Maximum value of $f(x)$ is attained at $x = \frac{\pi}{3}$

(D) Minimum value of $f(x)$ is attained at $x = 0$

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A), (B) and (D) only
3. (C) and (D) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Section : Core Mathematics

Q.51 The area (in sq. units) of the region in the first quadrant bounded by $y = 3\sqrt{1 - x^2}$, $x \in [0, 1]$ and the x -axis is equal to

1. $\frac{\pi}{4}$
2. $\frac{3\pi}{4}$
3. $\frac{\pi}{2}$
4. $\frac{\pi}{6}$

Options

1. 1
2. 2
3. 3
4. 4

Q.52 If $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along the co-ordinate axes OX , OY and OZ respectively, then

- (A) $\hat{i} \times \hat{j} = \hat{k}$
- (B) $\hat{k} \times \hat{i} = -\hat{j}$
- (C) $\hat{j} \cdot \hat{j} = 1$
- (D) $\hat{j} \cdot \hat{k} = 0$

Choose the **\textbf{correct}** answer from the options given below:

1. (A) and (B) only
2. (B), (C) and (D) only
3. (B) and (D) only
4. (A), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.53 If a line makes angles α , β and γ with the positive directions of x -axis, y -axis and z -axis respectively, then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$ is equal to

1. -2
2. 1
3. 2
4. 3

Options

1. 1
2. 2
3. 3
4. 4

Q.54 A person can sell a maximum of 20 units of shirts and pants on which a profit of ₹40 is made on each shirt and a profit of ₹30 on each pant. A minimum of 2 shirts are being sold, while pants are sold at least 4 times as many as shirts. Then the maximum profit is:

1. ₹480
2. ₹560
3. ₹640
4. ₹600

Options

1. 1
2. 2
3. 3
4. 4

Q.55 For the differential equation $ydx - (x + 3y^2)dy = 0$, which of the following statements are **\textbf{true}**?

(A) It is a linear differential equation

(B) It is a homogenous differential equation

(C) Its general solution is $x = 3y^2 + Cy$: C is an arbitrary constant

(D) If $y(0) = 1$, then its particular solution is $x = 3y^2 - 1$

Choose the \textbf{correct} answer from the options given below:

1. (A) and (C) only
2. (A), (B) and (C) only
3. (B), (C) and (D) only
4. (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.56
$$\int \left(\frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} \right) dx =$$

(Given that c is an arbitrary constant)

1. $2 \sin x + 2x \cos \alpha + c$
2. $2 \cos x + 2x \sin \alpha + c$
3. $x \cos x + 2 \sin \alpha + c$
4. $x \cos x + x \sin \alpha + c$

Options 1. 1

2. 2

3. 3

4. 4

Q.57 The area of the region $\{(x, y) : x^2 + y^2 \leq 1 \leq x + y\}$ is

1. $\left(\frac{\pi}{2} - 1\right)$ sq.unit
2. $\frac{1}{2}(\pi - 1)$ sq.unit
3. $\frac{1}{2}\left(\frac{\pi}{2} - 1\right)$ sq.unit
4. $(\pi - 1)$ sq.unit

Options 1. 1

2. 2

3. 3

4. 4

Q.58 The system of equations $x + y - z = 1, 3x + y - 2z = 3, x - y + \lambda z = 1$ has infinite number of solutions if λ is equal to

1. 0
2. 2
3. $-5/3$
4. any real number

Options 1. 1

2. 2

3. 3

4. 4

Q.59 Consider the line $\vec{r} = -2\hat{i} + 3\hat{j} + \hat{k} + \lambda(5\hat{i} - 3\hat{j} - \hat{k})$. Match \textbf{List-I} with \textbf{List-II}

List-I

(A) A point on the given line

(B) Direction ratios of the given line

(C) Direction cosines of the given line

(D) Direction ratios of a line perpendicular to given line

List-II

(I) $\left(\frac{5}{\sqrt{35}}, \frac{-3}{\sqrt{35}}, \frac{-1}{\sqrt{35}}\right)$

(II) (2, 3, 1)

(III) (5, -3, -1)

(IV) (-2, 3, 1)

Choose the \textbf{correct} answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
2. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
3. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
4. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.60 The relation R in the set $\{1, 2, 3\}$ given by $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (1, 3), (2, 3)\}$ is:

1. symmetric
2. not reflexive
3. not transitive
4. both reflexive and transitive

Options 1. 1

2. 2
3. 3
4. 4

Q.61 The product of order and degree of the differential equation $\left(\frac{d^3y}{dx^3}\right)^2 + x^2y\left(\frac{d^2y}{dx^2}\right)^3 = 2x^5$ is:

1. 3
2. 6
3. 2
4. 9

Options 1. 1

2. 2
3. 3
4. 4

Q.62 If A and B are invertible matrices of order 3 then match \textbf{List-I} with \textbf{List-II}

- | List-I | List-II |
|-----------------------|--------------------|
| (A) $\text{adj}(A)$ | (I) $B^{-1}A^{-1}$ |
| (B) $(AB)^{-1}$ | (II) $ A ^{-1}$ |
| (C) $ A^{-1} $ | (III) $ A ^2$ |
| (D) $ \text{adj } A $ | (IV) $ A A^{-1}$ |

Choose the \textbf{correct} answer from the options given below:

1. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
2. (A) - (IV), (B) - (I), (C) - (III), (D) - (II)
3. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
4. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.63 Match \textbf{List-I} with \textbf{List-II}

- | \textbf{List-I} | \textbf{List-II} |
|-------------------------------------|------------------|
| (A) $\int_0^1 \frac{x^2}{1+x^3} dx$ | (I) 0 |

(B) $\int_0^{\pi} 3 \sin x dx$ (II) $2 \log_e \left(\frac{3}{2} \right)$

(C) $\int_{-1}^1 \sin^5 x \cos^6 x dx$ (III) 6

(D) $\int_2^3 \frac{4}{x^2 - 1} dx$ (IV) $\frac{1}{3} \log_e 2$

Choose the \textbf{correct} answer from the options given below:

1. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
2. (A) - (IV), (B) - (I), (C) - (III), (D) - (II)
3. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
4. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.64 The rate of change of area of a circle with respect to its circumference when radius is 6 cm, is

1. $2 \text{ cm}^2/\text{cm}$
2. $4 \text{ cm}^2/\text{cm}$
3. $6 \text{ cm}^2/\text{cm}$
4. $12 \text{ cm}^2/\text{cm}$

Options 1. 1

2. 2

3. 3

4. 4

Q.65 If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and $|\vec{a}| = 5$, $|\vec{b}| = 3$, $|\vec{c}| = 7$, then the acute angle between $\vec{a}$ and $\vec{b}$ is

1. $\frac{\pi}{4}$
2. $\frac{\pi}{2}$
3. $\frac{\pi}{6}$
4. $\frac{\pi}{3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.66 Let $A = [a_{ij}]_{2 \times 4}$ and $B = [b_{ij}]_{4 \times 2}$, then $|3AB|$ is equal to

1. $3^2|AB|$
2. $3^4|AB|$
3. $3^2|A||B|$
4. $3^4|A||B|$

Options 1. 1

2. 2

3. 3

4. 4

Q.67 If the points P, Q, R with position vectors $5\hat{i} + \lambda\hat{j}$, $20\hat{i} - \hat{j}$ and $15\hat{i} - 6\hat{j}$ respectively are collinear, then the value of λ is

1. -16
2. 6
3. 5
4. 12

Options 1. 1

2. 2

3. 3

4. 4

Q.68 Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I}

\textbf{List-II}

(A) The maximum value of $f(x) = \sin(3x) + 6$ (I) 2

(B) The maximum value of $f(x) = -|x + 2| + 4$ (II) 5

(C) The minimum value of $f(x) = (3x + 1)^2 + 5$ (III) 7

(D) The minimum value of $f(x) = 2 \cos x + 4$ (IV) 4

Choose the \textbf{correct} answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

3. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

4. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.69 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = 100x + 1$, where $\mathbb{R}$ is a set of real numbers, then

1. f is one-one but not onto

2. f is onto but not one-one

3. f is both one-one and onto

4. f is neither one-one nor onto

Options 1. 1

2. 2

3. 3

4. 4

Q.70 A die is tossed once. If the random variable X is defined as $X = \begin{cases} 1, & \text{if the die result in an odd number} \\ -1, & \text{if the die result in an even number} \end{cases}$, then the variance of X is

1. 0

2. 1

3. $\frac{1}{2}$

4. $\frac{1}{4}$

Options 1. 1

2. 2

3. 3

4. 4

Q.71 The largest open interval, in which the function $f(x) = \frac{x}{x^2 + 1}$ increases, is

1. $(0, 1)$

2. $(-1, 0)$

3. $(-1, 1)$

4. $(-\infty, -1) \cup (1, \infty)$

Options 1. 1

2. 2

3. 3

4. 4

Q.72 The shortest distance between the lines $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$ and $\vec{r} = (2\hat{i} + 4\hat{j} + 5\hat{k}) + \mu(4\hat{i} + 6\hat{j} + 8\hat{k})$ is equal to

1. $\sqrt{5}$
2. $\sqrt{29}$
3. $\sqrt{\frac{29}{5}}$
4. $\sqrt{\frac{5}{29}}$

Options

1. 1
2. 2
3. 3
4. 4

Q.73

Let $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, then $(A^{-1})^T$ equals

1. $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$
2. $\begin{bmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$
3. $\begin{bmatrix} -\cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$
4. $\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$

Options

1. 1
2. 2
3. 3
4. 4

Q.74

If A and B are skew-symmetric matrices, then which of the following is **not** true?

1. $A^5 + B^7$ is skew-symmetric
2. A^{21} is skew-symmetric
3. B^{18} is symmetric
4. $A^6 + B^7$ is symmetric

Options

1. 1
2. 2
3. 3
4. 4

Q.75

Let $y = \cos(\sin x^2)$, then the value of $\frac{dy}{dx}$ at $x = \frac{\sqrt{\pi}}{2}$ is equal to

1. $-\sqrt{\frac{\pi}{2}} \sin\left(\frac{1}{\sqrt{2}}\right)$
2. $-\frac{\sqrt{\pi}}{2} \sin\left(\frac{1}{\sqrt{2}}\right)$
3. $-\sqrt{\frac{\pi}{2}} \cos\left(\frac{1}{\sqrt{2}}\right)$
4. $\sqrt{\frac{\pi}{2}} \cos\left(\frac{1}{\sqrt{2}}\right)$

Options

1. 1
2. 2
3. 3
4. 4

Q.76

Match **List-I** with **List-II**

Let A and B be two events such that $P(A) = 0.2$, $P(B) = 0.4$, $P(B|A) = 0.5$

List-I **List-II**

(A) $P(A \cap B)$ (I) 0.5

(B) $P(A|B)$ (II) 0.8

(C) $P(A \cup B)$ (III) 0.25

(D) $P(A')$ (IV) 0.1

Choose the \textbf{correct} answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
2. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (II), (B) - (IV), (C) - (III), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.77

The probability that A hits a target is $\frac{1}{5}$ and the probability that B hits it is $\frac{2}{3}$. The probability that the target will be hit if both A and B shoot at it independently is:

1. $\frac{11}{15}$
2. $\frac{1}{15}$
3. $\frac{2}{15}$
4. $\frac{4}{15}$

Options 1. 1

2. 2

3. 3

4. 4

Q.78

Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I} \textbf{List-II}

(A) $f(x) = |x| + 1$ (I) Not differentiable at $x = 3$ only

(B) $f(x) = |x - 3|$ (II) Not differentiable at $x = -3$ only

(C) $f(x) = |x + 3|$ (III) Not differentiable at $x = 3, -3$ only

(D) $f(x) = |x^2 - 9|$ (IV) Not differentiable at $x = 0$ only

Choose the \textbf{correct} answer from the options given below:

1. (A) - (I), (B) - (III), (C) - (IV), (D) - (II)
2. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)
3. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
4. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Options 1. 1

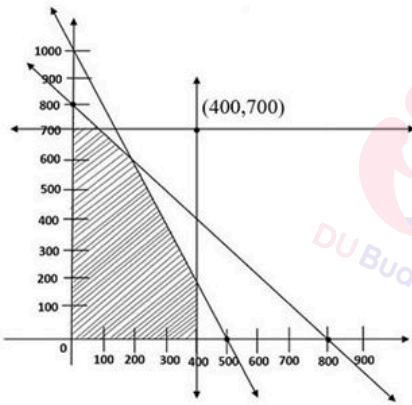
2. 2

3. 3

4. 4

Q.79

Which one of the following set of constraints represents the shaded region given below?



1. $x + y \leq 800, 2x + y \leq 1000, x \leq 400, y \leq 700, x \geq 0, y \geq 0$
2. $x + y \geq 800, 2x + y \leq 1000, x \leq 400, y \geq 700, x \geq 0, y \geq 0$
3. $x + y \leq 800, 2x + y \geq 1000, x \geq 400, y \leq 700, x, y \geq 0$
4. $x + y \geq 800, 2x + y \geq 1000, x \leq 400, y \leq 700, x \geq 0, y \geq 0$

Options 1. 1

2. 2

3. 3

4. 4

Q.80

Let $A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & -1 \\ 4 & 1 & 2 \end{bmatrix}$. M_{ij} and A_{ij} respectively denote the minor and cofactor of an element a_{ij} of matrix $A = [a_{ij}]$.

(A) $M_{23} = 6$

(B) $A_{22} = -8$

(C) $A_{13} = 7$

(D) $M_{32} = -5$

Choose the \textbf{correct} answer from the options given below:

1. (A), (C) and (D) only

2. (B) and (D) only

3. (C) and (D) only

4. (A), (B) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.81

'A' speaks the truth in 80\% of the cases while 'B' in 90\% of the cases. The probability that they contradict each other in stating the same statement is

1. $\frac{37}{50}$

2. $\frac{13}{50}$

3. $\frac{18}{25}$

4. $\frac{9}{25}$

Options 1. 1

2. 2

3. 3

4. 4

Q.82

$\cos^{-1} \left(\cos \frac{7\pi}{6} \right)$ equals:

1. $\frac{7\pi}{6}$
2. $\frac{5\pi}{6}$
3. $-\frac{5\pi}{6}$
4. $\frac{\pi}{6}$

Options

1. 1
2. 2
3. 3
4. 4

Q.83

Let $\vec{a} = 2\hat{i} - \hat{j}$, $\vec{b} = -4\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + 2\hat{k}$. If $\vec{d}$ is a vector perpendicular to both $\vec{a}$ and $\vec{b}$ such that $\vec{c} \cdot \vec{d} = 34$, then $|\vec{d}|$ is equal to

1. $\sqrt{69}$
2. $2\sqrt{69}$
3. $3\sqrt{69}$
4. $4\sqrt{69}$

Options

1. 1
2. 2
3. 3
4. 4

Q.84

$\int_0^{\pi/2} \frac{\sin^8 x}{\sin^8 x + \cos^8 x} dx$ is equal to

1. $\frac{\pi}{2}$
2. $\frac{\pi}{4}$
3. 0
4. $-\frac{\pi}{2}$

Options

1. 1
2. 2
3. 3
4. 4

Q.85

For what value of α , the function f defined by $f(x) = \begin{cases} \alpha(x^2 - 2x + 1), & \text{if } x \leq 0 \\ 2x + 1, & \text{if } x > 0 \end{cases}$ is continuous at $x = 0$?

1. $\alpha = 1$
2. $\alpha = 2$
3. $\alpha = -1$
4. $\alpha = 0$

Options

1. 1
2. 2
3. 3
4. 4